

Project Proposal: Paralleling Human Neuroplasticity through Transformers

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Pruning models to compress their size while retaining accuracy has been an active area of exploration in ML. This project aims to study how pruning can be used to enhance transformer accuracy in continual learning by replicating human neuroplasticity. Neuroplasticity refers to the brain's ability to change its neural connectivities from experiences, including strengthening and weakening different neural pathways. Our hypothesis is that pruning can be a tool to model aspects of forgetting within transformers and improve overall accuracy. Pruning and resetting allow the model to strengthen important connections and develop more relevant connections. We will explore how different structures of forgetting influence model accuracy over continual learning tasks.

This methodology is inspired by the methodology described in the TriRe framework (12). We will use the existing BERT-tiny transformer and introduce a sequence of 5 classification tasks to it. For each task, a large majority of the data will be task-specific, and a small portion will act as memory to reduce catastrophic forgetting. For every task, we first train the model on the combined data, then use Continual Weight Importance to identify “important” weights. The CWI is based on their magnitude and sensitivity to current and previous losses. We will partition the important and unimportant weights. The first checkpoint is a model with unimportant weights pruned. Next, the important weights are fine-tuned on the task specific data and the unimportant weights are trained on the memory data. After allowing these weights to train separately, they are merged. This is the second checkpoint. Next we go back to the first checkpoint and reset any non-important weights to their initial value and retrain on the full-dataset. This parallels forgetting unused information to make space for new connections. This is the third checkpoint. The same process will be applied to the model created at the second checkpoint, making the fourth checkpoint. We will then compare the accuracies of the 4 checkpoints on the test dataset to analyze the effects of structured pruning, selective forgetting, and retraining strategies in continual learning.

We plan to use HuggingFace’s pretrained BERT-tiny model. We will implement CWI scoring and the pipeline using PyTorch, where we can use gradient hooks to zero out weights to smoothly partition and merge across checkpoints (11). We plan to use 5 types of classification data: SST-2 (sentiment analysis), AG-News (topic), TREC (question type), DBpedia-14 (text classification), and Yahoo Answers (question topic), all publicly available through HuggingFace datasets. To ensure an equal representation, we will use 5,000 training samples per task, 500 of those samples will be randomly selected to be included in the memory dataset. For our test dataset, we will use 500 samples per task. The first milestone will be creating a model that’s able to go through 2 learning tasks, completing only the first checkpoint.

This project is computationally feasible because we chose to use a lightweight model (4.4 million parameters, 2 hidden layers), designed to have faster inference and fine-tuning times (13). We also choose a relatively light dataset to process, with a maximum of 5500 samples being trained per checkpoint which will run on Google Colab Pro’s A100 GPU. In an initial training run of BERT-tiny over 5,500 datapoints, each epoch took 1.9s. This means over our 5 tasks and 25 training runs total we estimate around 30 minutes total for the full pipeline, making it highly feasible within Colab limits.

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